Kit description:

This kit helps students explore how electricity can be created from movement — just like real wind turbines that turn the wind's power into clean energy. By learning how a simple motor can generate electricity, students see how renewable energy works in action. Green energy, such as wind and solar power, is important because it reduces pollution, protects the planet, and helps build a sustainable future. Every small experiment like this shows how science and creativity can work together to power the world responsibly.

Scenario

Imagine a small rural village far from the city. The people there don't have access to reliable electricity — when the sun goes down, homes become dark, and students can't study at night. Fuel-powered generators are expensive and create pollution, so the community needs a cleaner, more sustainable solution.

Your mission is to design a small wind turbine that can turn wind energy into electricity using simple materials. By experimenting with your motor and understanding how movement can generate power, you will explore how renewable energy could bring light and opportunity to communities that need it most.

Safety

- Do NOT shorten the 9V battery terminals together; this can cause heat and damage.
- LEDs are bright—avoid staring directly into the light.
- Keep small parts away from mouths as it can be a choking hazard.
- Keep long hair, fingers, and loose clothing away from any spinning or moving parts.
- Disconnect the battery when not in use to prevent overheating.

Vocabulary

- **Circuit:** A closed loop path for electricity.
- **Series:** Components in a single path; current flows through each in turn.
- **Polarity:** Direction-sensitive parts (battery, LED) have a + and – side.
- **Resistor:** Limits current, so components don't burn out.
- **Switch:** Opens (off) or closes (on) the circuit.
- **Generator:** A device that converts mechanical energy (motion) into electrical energy.
- **Mechanical Energy:** Energy from movement or motion.
- **Electrical Energy:** Energy from the flow of electric current.
- **Motor:** A device that converts electrical energy into mechanical motion.
- **Switch:** Opens (off) or closes (on) the circuit.
- **Power Source:** Provides electrical energy to run the circuit.

LED lesson

Purpose

Students build a simple flashlight to learn about **power sources, polarity, buttons, resistors**, and **basic circuit flow**.

Learning Objectives

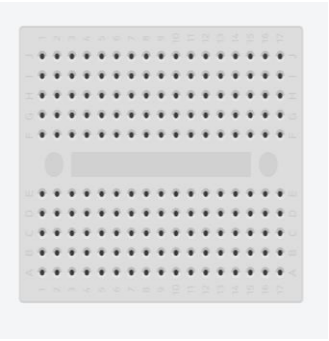
- Identify battery **positive/negative** and LED **anode/cathode**.
- Explain why an LED needs a **resistor** in series with a battery.
- Assemble a **series of circuits** with a button that controls an LED.

Materials

- 1× 9V battery **with snap connector leads**
- 1× LED (red recommended for first build)
- 1× resistor **470 Ω** (acceptable: 330–1k Ω)
- 1× push button **or** toggle switch (SPST)
- ~6× jumper wires (male–male for breadboard)
- 1× breadboard (400–600 tie points recommended)
- Optional: tape, small cardboard tube (to make a “flashlight body”), marker labels

Build Steps

What is a **breadboard**?
A breadboard is a small plastic board used to build and test electrical circuits without soldering. It has many little holes where you can plug in wires and components like LEDs or resistors. Inside the board, metal strips connect certain holes together so electricity can flow between them. It’s like a reusable puzzle that lets you easily create, test, and change circuits.



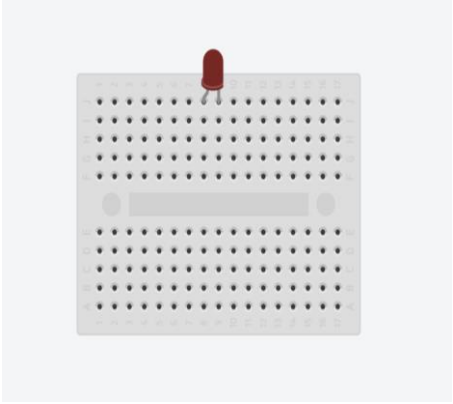
1. **Prepare battery**:
   Connect the 9V snap to the battery (red = +, black = −). Keep battery disconnected from the circuit while building.

2. **Identify LED legs**:

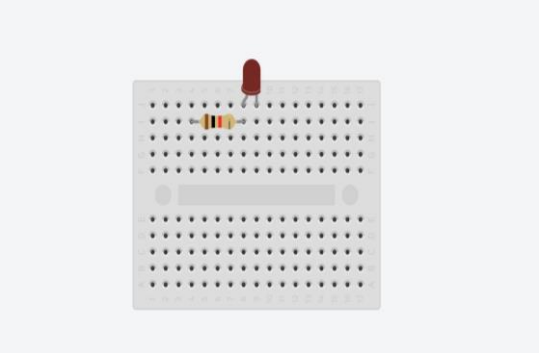
   a. **Long leg = anode (+)**.
   b. **Short leg = cathode (−)**.
      If legs are trimmed and you can't tell, the flat side of the LED trim is the cathode.



3. **Place LED on the breadboard**:
   a. Insert LED so each leg sits in a *different* row (not the same row). — The point: each leg must be in a separate electrically connected row.



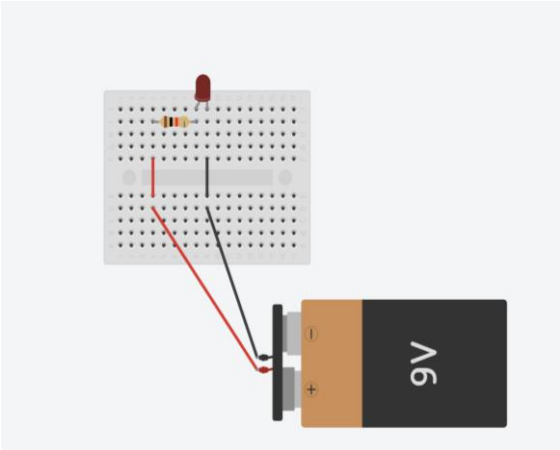
4. **Add the resistor:**
    a. Connect one end of the resistor into the **same row** as the LED's **anode (long leg)**.
    b. Put the other end of the resistor into a different row (this will be the row we connect to the battery +).



5. **Wire the battery + to the resistor:**
    Insert the positive jumper cable into the same row as the resistor.

6. **Connect LED cathode to battery:**
    Insert the batteries negative (black) jumper wire into the same row as the LED's cathode.

7. **Final check:**
    a. Verify resistor is in series with the LED anode; LED cathode goes to negative, no short circuits (no red directly to black).
    b. Ensure LED orientation: long leg → resistor → +9V; short leg → ground.



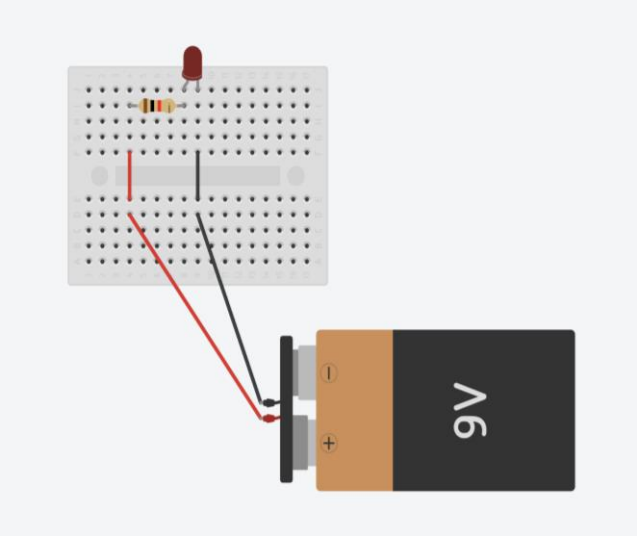
8. **Connect the battery:**
    Plug the 9V battery into the snap. LED should light.

9. **Troubleshooting:**
    a. LED reversed? Flip LED legs.
    b. Loose connections? Re-seat wires/parts.
    c. Bad battery? Measure or replace — a weak 9V may not light LED light brightly.
    d. LED burnt (if no resistor was used) — replace LED and use resistor.

Adding a button

Review the original LED setup

From your previous LED circuit:

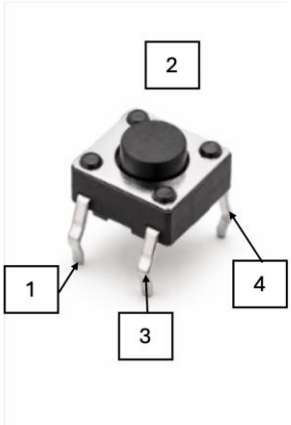


Now we will add a **button** so the LED only lights when pressed.

2. Understand how the button works

A typical **tactile switch** has **four pins** — but they are two **pairs of connected pins**:

[1] [2]
[3] [4]



- Pins **1–2** are connected together.
- Pins **3–4** are connected together.
- When you **press** the button, **the two sides connect** (1–2 connect to 3–4).

So you will connect one side of the button to the battery's positive voltage, and the other side to the LED circuit.

(If your button has just 2 pins, it's even simpler — they connect when pressed.)

3. Place the button on the breadboard

- Push the button so that its **legs straddle the center gap** of the breadboard.
  This ensures that each side of the button is on a separate circuit row.

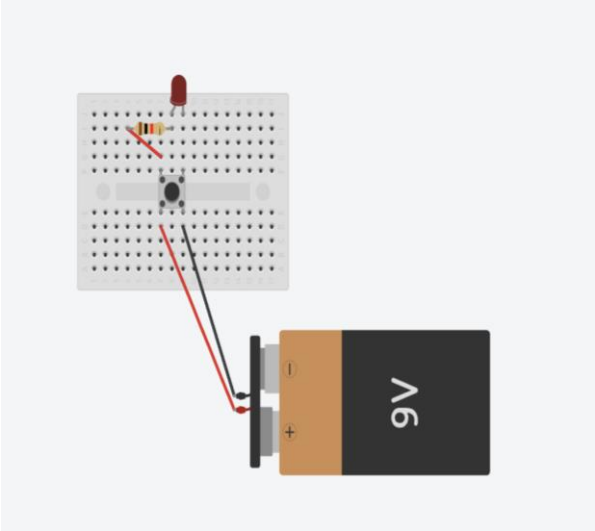
4. Re-wire the circuit with the button in series

Now wire it like this:

1. **Battery positive (red wire)** → goes to **one side of the button** (one row).
2. **The other side of the button** → connect a **jumper wire** from that row to **one end of the resistor**.
3. **Resistor** → connect to **LED anode (long leg)**.
4. **LED cathode (short leg)** → connect to **battery negative (black wire)**.

Visually simplified:

9V(+) → [Button] → [Resistor] → [LED →|] → 9V(−)



5. Double-check

- Button straddles center gap.
- The resistor is still in series with the LED.
- Long leg (anode) of LED connects toward + side.
- No loose or touching wires.

6. Connect the battery

- Attach the 9 V snap to the battery.
- Press the button — **the LED should light up!**
- Release it — **the LED turns off.**

Motor lesson

Purpose

Students will build a simple fan circuit using a DC motor to learn about power sources, polarity, switches, and how electrical energy can be converted into motion.

Learning Objectives

- Identify the positive and negative sides of a battery and motor.
- Explain how a DC motor converts electrical energy into mechanical energy.
- Assemble a simple circuit with a switch that controls the motor.
- Observe how reversing the polarity changes the motor's spin direction.

Materials

- 1× 9V battery with snap connector leads
- 1× small DC motor (3–9V range)
- 1× push button or toggle switch (SPST)
- ~6× jumper wires (male–male for breadboard)
- 1× breadboard (400–600 tie points recommended)
- Optional: small paper or plastic fan blade to attach to motor shaft, tape, marker labels

Build Steps

1. Identify motor terminals

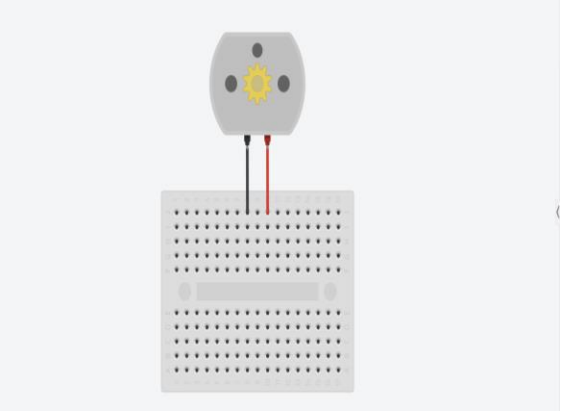
Your DC motor has **two metal terminals**.

- One terminal = positive (+)

- The other = negative (−)
  If you reverse them, the motor spins the **opposite direction** — so polarity just controls spin direction.

2. Place the motor

Insert each of the motor's two terminals into **different rows** of the breadboard
Make sure the motor is secure and the terminals don't touch each other



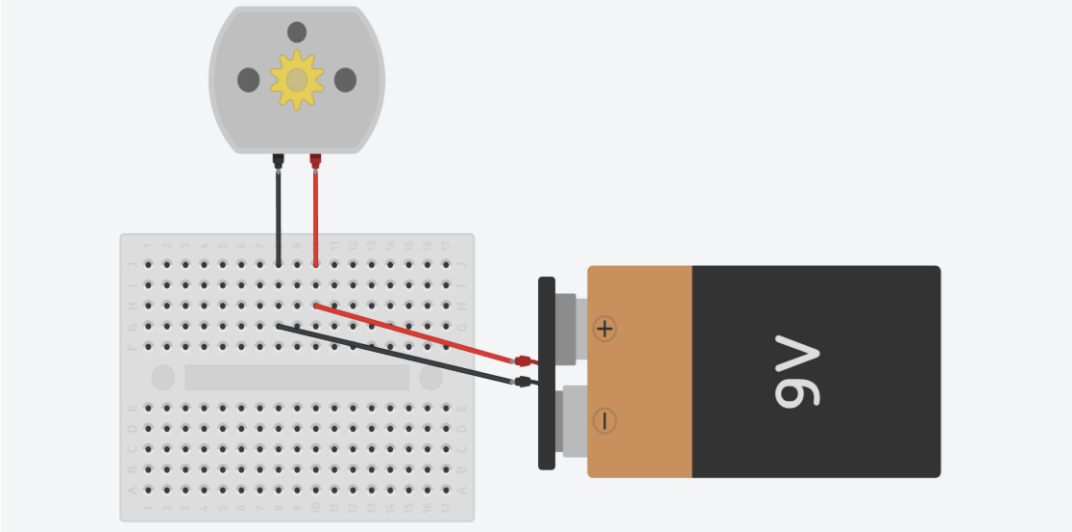
3. Connect the battery

Use the **battery snap** wires:

- **Red wire (positive)** → insert into the same row as the motor's **positive terminal**.
- **Black wire (negative)** → insert into the same row as the motor's **negative terminal**.

4. Connect the battery

Snap the 9 V battery into the clip.



Optional: Reverse direction! (for fun)

To make the motor spin the opposite direction:
Swap the battery wires on the breadboard — red ↔ black (reverse polarity).

Challenge: Can you add a button to the motor to control when it spins?

Wind turbine

Purpose

Students will use a DC motor as a simple generator to learn how mechanical energy can be converted into electrical energy. They will see how spinning the motor's shaft produces electricity that can power a small load, such as an LED.

Learning Objectives

- Explain how a DC motor can also work as a generator.
- Describe the relationship between motion (mechanical energy) and electricity (electrical energy).
- Build a simple circuit where a motor generates power to light an LED.
- Observe how spinning the motor faster increases voltage and brightness.

Materials You'll Need

- 1 × **Small DC motor** (3–9 V type works great — same as before)
- 1 × **LED** (red LED works best; it lights at ~2 V, lower than blue/white)
- 1 × **Breadboard**
- 2 × **Jumper wires**
- **Fan blade, propeller, or small wheel** to spin the motor shaft

Step-by-Step Instructions

1. Understand what's happening

A **DC motor** and a **generator** are the same device used in reverse.

- When you **apply voltage**, the motor spins.
- When you **spin the shaft**, it *produces* voltage and current.

This generated electricity can light an LED — if you spin it fast enough!

2. Prepare your LED

- Identify LED legs:
  - **Long leg = positive (anode)**
  - **Short leg = negative (cathode)**

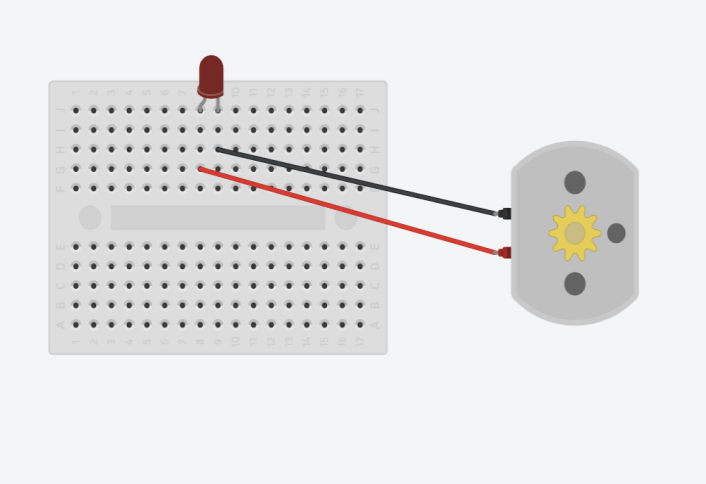
If using a breadboard, insert the LED so each leg is in a separate row.

3. Connect the motor to the LED

Use jumper wires or alligator clips:

- **Motor terminal A → LED anode (long leg)**
- **Motor terminal B → LED cathode (short leg)**

That's it — no battery! The motor itself will generate the power.



5. Spin the motor

- Spin the shaft **quickly** by blowing on the fan.
- The LED should **flash or glow** faintly at first.

If it does not light up, reverse the LED leads — the polarity may be flipped.